

VAPO: Value-Model-Based RL for Advanced Reasoning

Achieving State-of-the-Art on AIME 2024

60.4

AIME 2024 (avg@32)
Surpassing DAPO by 10+ points

Base Model:	Qwen2.5-32B (no SFT data)
Training:	~5,000 steps, zero crashes
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Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO / DAPO)

- ✗ Advantage computed from full trajectory reward
- ✗ Reward assigned uniformly across all tokens
- ✗ Cannot distinguish contribution of individual steps
- ✗ High variance on long reasoning chains
- ✗ Subtle step-level errors may go unpunished



Value-Model-Based (VAPO)

- ✓ Token-level credit assignment via value model
- ✓ Traces each action's impact on final return
- ✓ Critical when subtle step errors cause failures
- ✓ Lower variance per-token vs Monte Carlo
- ✓ Value model generalises → better sample efficiency

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

01

Value Model Bias

Long trajectories accumulate bootstrapped estimation errors. Each step's value estimate compounds upstream error, causing the value model to drift from ground truth — especially during early training when the model is least reliable.

02

Heterogeneous Sequence Lengths

Reasoning traces vary from a few hundred to tens-of-thousands of tokens. A fixed λ in GAE imposes the same bias-variance tradeoff on every sequence length, which is sub-optimal: short responses need lower λ (lower variance) while long ones need higher λ (lower bias).

03

Reward Signal Sparsity

Only the final `<eos>` token receives a binary reward signal. With long CoT the credit must propagate back through thousands of tokens, creating extremely sparse learning signal and making it harder for the value model to assign meaningful per-step credit.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenge Addressed	VAPO Component	Source	Description
Value Model Bias	Value-Pretraining	Novel	Pre-train value model before RL to reduce initial estimation bias
	Decoupled-GAE	VC-PPO	Separate GAE computation for more robust value estimation
Heterogeneous Seq Lengths	Length-Adaptive GAE	Novel	Dynamically adjust λ per sequence length to balance bias-variance
Reward Sparsity / Explore	Clip-Higher	DAPO	Asymmetric clipping encourages exploration of new reasoning paths
Training Stability	Token-level Loss	DAPO	Normalize loss at token level rather than sequence level
	Positive Example LM Loss	Novel	Self-imitation on correct trajectories reinforces good reasoning chains
Baseline Estimation	Group-Sampling	GRPO	Sample multiple responses per prompt for better advantage baseline

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Novel component that resolves the heterogeneous sequence length challenge

The Core Idea

In standard GAE, the λ parameter controls the bias-variance tradeoff uniformly across all sequences.

VAPO's Length-Adaptive GAE makes λ a function of response length:

- Short responses $\rightarrow$ Lower λ
(prioritise lower variance; fewer steps, less need for long-horizon returns)
- Long responses $\rightarrow$ Higher λ
(prioritise lower bias; more steps, bootstrapping error accumulates $\rightarrow$ need longer-horizon estimates)

This per-sequence adaptation ensures each reasoning trace receives the optimal bias-variance balance during training.

Short Response (few hundred tokens)

$\lambda \rightarrow \text{LOW}$

Lower variance | TD-style advantage | Less sensitivity to value errors

Bootstrapping over a short horizon keeps variance low and value estimation errors have less time to accumulate.

Long Response (thousands of tokens)

$\lambda \rightarrow \text{HIGH}$

Lower bias | Monte-Carlo-style advantage | Long-horizon credit

Longer returns reduce bias so early reasoning steps receive accurate credit even when final reward is thousands of tokens away.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO & DeepSeek-R1-Zero

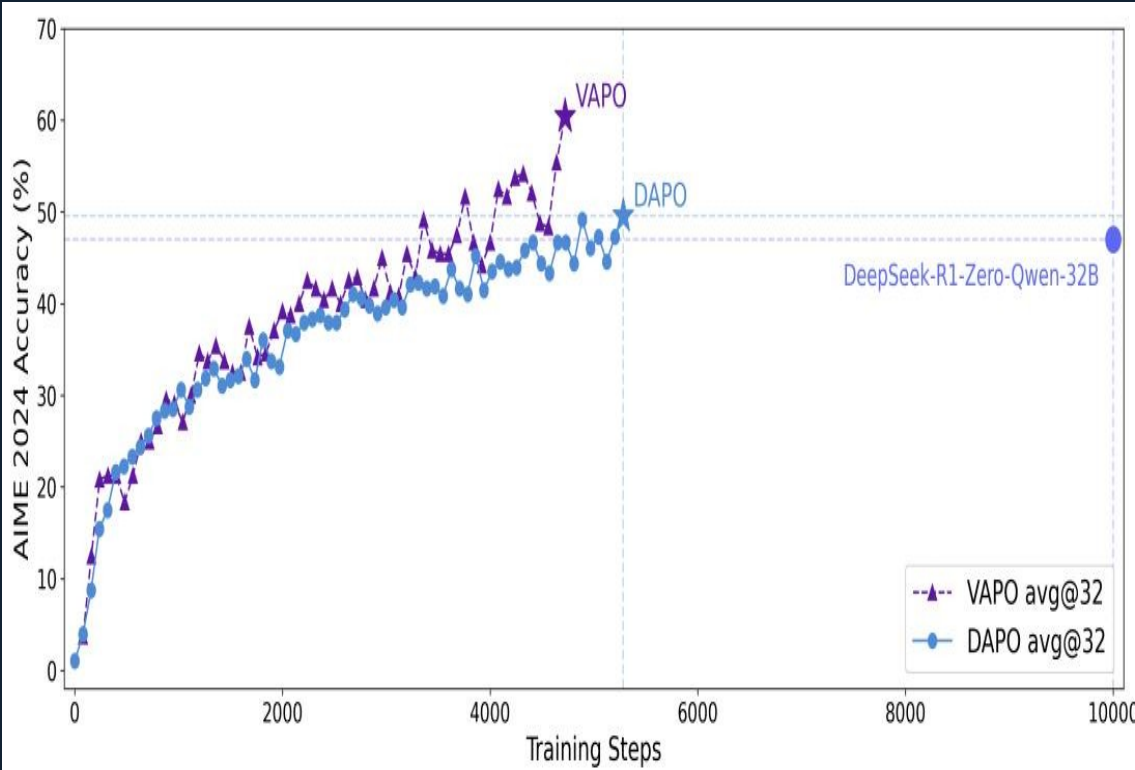
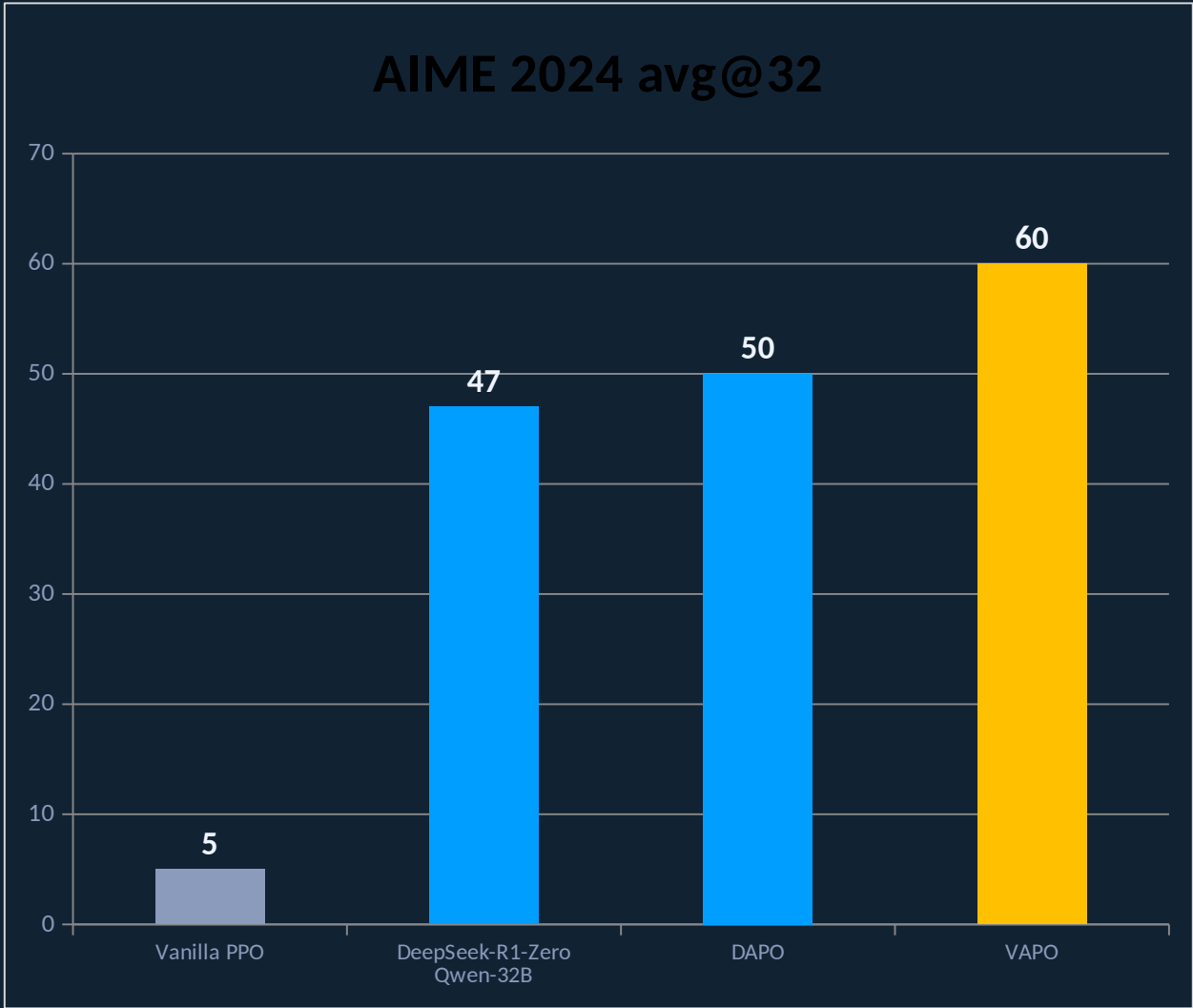


Figure 2: VAPO vs DAPO training curves on AIME 2024. VAPO reaches 60% at ~4,500 steps; DAPO plateaus at 50% at ~5,500 steps.

VAPO (60) > DAPO (50) > DeepSeek-R1-Zero (47) > Vanilla PPO (5) | All on Qwen2.5-32B, no SFT

VAPO Reaches SOTA in 5K Steps — With Superior Stability and Longer Reasoning

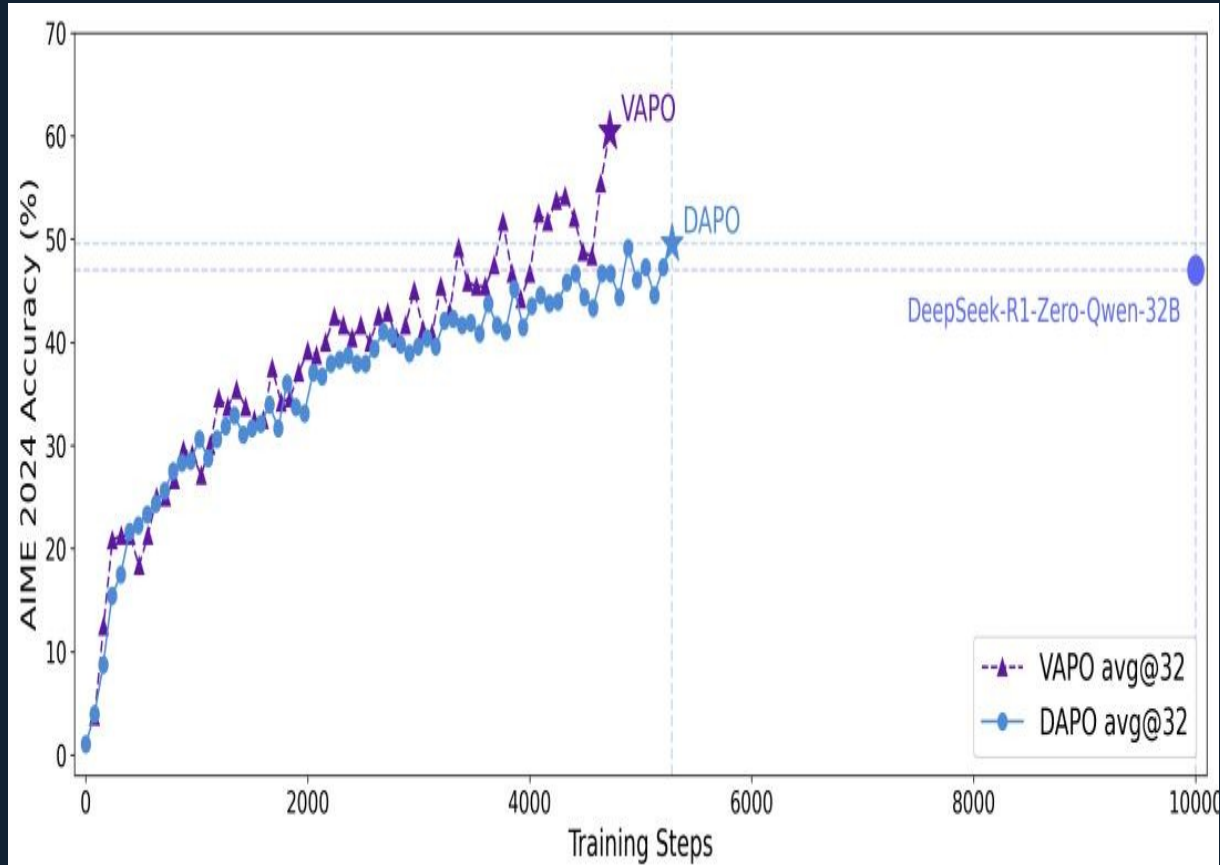


Figure 2: Accuracy over training steps
VAPO ≈60% at ~4,500 steps | DAPO ≈50% at ~5,500 steps
DeepSeek-R1-Zero ≈47% at ~10,000 steps

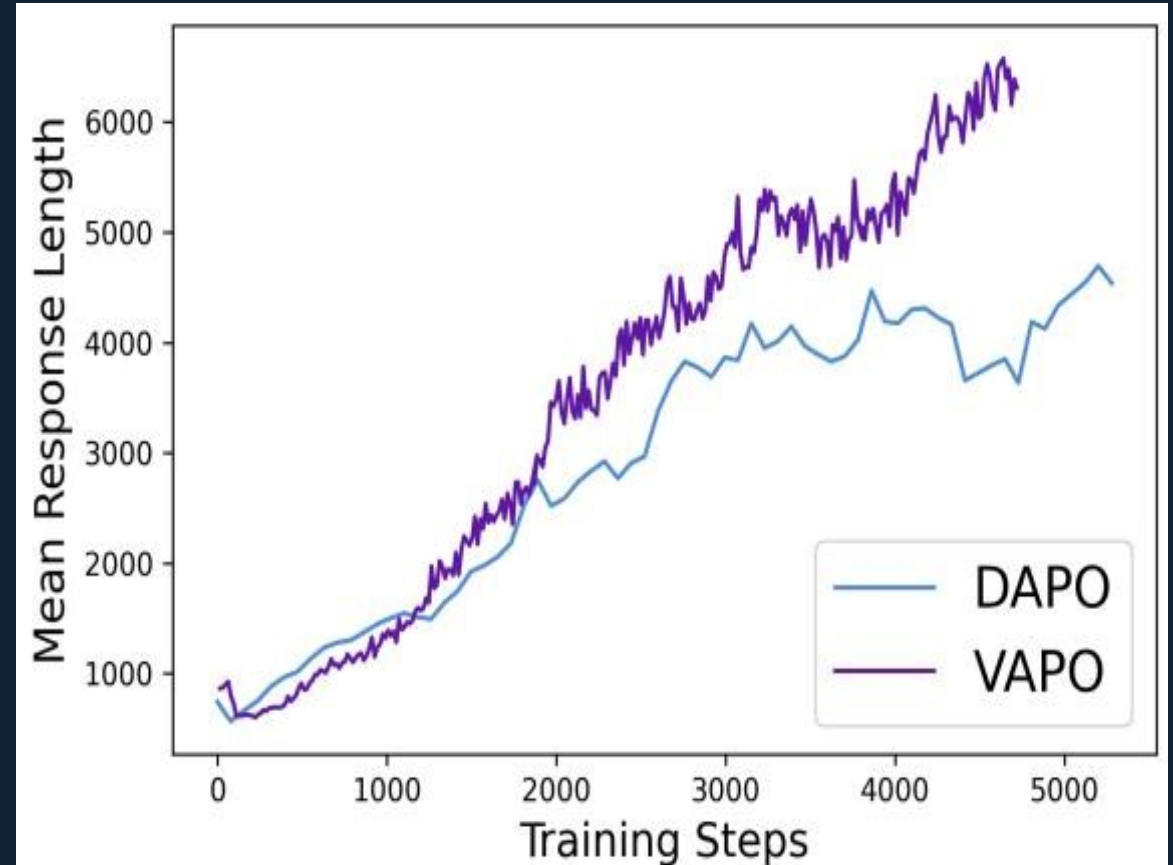


Figure 3: Mean response length over training
VAPO reaches ~6,000 tokens | DAPO reaches ~4,500 tokens
Longer chains → more elaborate reasoning

~5,000 steps

VAPO reaches SOTA

~5,500 steps

DAPO, lower score

0 crashes

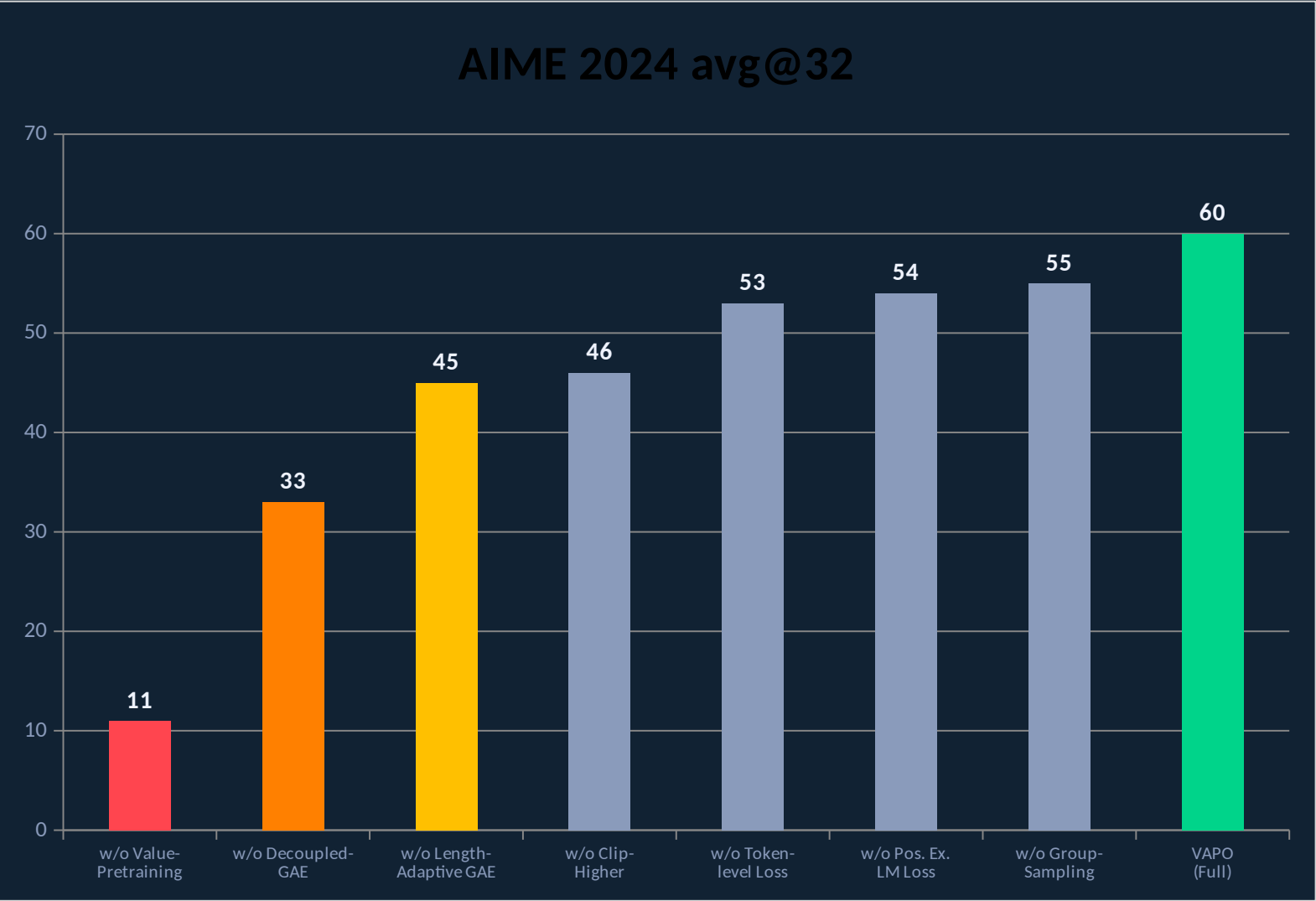
across multiple runs

~6,000 tok

VAPO mean response

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

AIME 2024 avg@32



Remove Component → Score Drop

Component	Score	Drop	Impact
Value-Pretraining	60 → 11	-49 pts	CRITICAL
Decoupled-GAE	60 → 33	-27 pts	SEVERE
Length-Adaptive GAE	60 → 45	-15 pts	HIGH
Clip-Higher	60 → 46	-14 pts	HIGH
Token-level Loss	60 → 53	-7 pts	MED
Pos. Example LM Loss	60 → 54	-6 pts	MED
Group-Sampling	60 → 55	-5 pts	MED

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based RL can surpass value-free methods

VAPO (60) beats DAPO (50) and DeepSeek-R1-Zero (47) on AIME 2024 using the same Qwen2.5-32B base — proving that, when three core challenges are systematically addressed, value-based approaches regain their theoretical edge.

2

Value-Pretraining is the single most important ingredient

Removing it collapses performance from 60 $\rightarrow$ 11 (~49 pts). Without a warm-started value model, value-based RL barely functions, confirming that initial bias is the dominant failure mode.

3

Length-Adaptive GAE solves a novel problem unique to long-CoT

Dynamic λ adjustment by sequence length contributes 15 points. Fixed- λ GAE cannot handle the heterogeneous lengths that emerge during reasoning model training.

4

VAPO is stable and training-efficient

Zero crashes across multiple independent runs; SOTA reached in ~5,000 steps vs DAPO's ~5,500 for a lower score. Stability is a first-class property, not a side-effect.

5

Implications for future reasoning model development

Value-model quality, bias initialisation, and per-sequence bias-variance tuning are key levers. Future work should explore scaling value models alongside policy models and extending Length-Adaptive GAE to other reasoning domains.